Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
AI and Application Development and Jira

Task 1

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Q.1 **In a college:**

- 20% of students know Python.
- 80% of students know Java.
- 90% of Python students get placed.
- 40% of Java students get placed.

A randomly selected student is placed.Find:

- Probability that the student knows Python.
- Probability that the student knows Java.
- Probability that a placed student knows Python.

Code:

```
# Probability that a student knows Python
P_Python = 0.20

# Probability that a student knows Java
P_Java = 0.80

# Probability of getting placed if the student knows Python
P_Placed_Python = 0.90

# Probability of getting placed if the student knows Java
P_Placed_Java = 0.40

# Probability that a randomly selected student is placed
P_Placed = (P_Python * P_Placed_Python) + (P_Java * P_Placed_Java)

# Probability that a placed student knows Python
P_Python_Placed = (P_Python * P_Placed_Python) / P_Placed

# Probability that a placed student knows Java
P_Java_Placed = (P_Java * P_Placed_Java) / P_Placed

print("Probability that the student knows Python:")
print(round(P_Python * 100, 2), "%")

print("Probability that the student knows Java:")
print(round(P_Java * 100, 2), "%")

print("Probability that a placed student knows Python:")
print(round(P_Python_Placed * 100, 2), "%")

print("Probability that a placed student knows Java:")
print(round(P_Java_Placed * 100, 2), "%")
```

```
print("Afsha 25")
```

Output:

```
Probability that the student knows Python:
20.0 %
Probability that the student knows Java:
80.0 %
Probability that a placed student knows Python:
36.0 %
Probability that a placed student knows Java:
64.0 %
Afsha 25
```

Q.2 **In a city:**

- 85% of taxis are Green.
- 15% are Blue.
- A witness correctly identifies the taxi color 80% of the time.

A witness claims the taxi involved in an accident was Blue. What is the probability that the taxi was actually Blue?

Code:

```
# Probability that the taxi is Green
P_Green = 0.85

# Probability that the taxi is Blue
P_Blue = 0.15

# Probability that witness correctly identifies the color
P_Correct = 0.80

# Probability that witness is wrong
P_Wrong = 1 - P_Correct

# Probability that witness claims Blue if taxi is actually Blue
P_Claim_Blue_Blue = P_Correct

# Probability that witness claims Blue if taxi is actually Green
P_Claim_Blue_Green = P_Wrong

# Bayes Rule
P_Blue_Claim = (P_Blue * P_Claim_Blue_Blue) / (
    (P_Blue * P_Claim_Blue_Blue) +
    (P_Green * P_Claim_Blue_Green)
)

print("Probability that the taxi was actually Blue:")
print(round(P_Blue_Claim * 100, 2), "%")
print("Afsha 25")
```

Output:

```
Probability that the taxi was actually Blue:
41.38 %
Afsha 25
```